

Cloud computing
 - store things on internet via internet storage space
 - bunch of computer around
 - cloud storage = 99.99%
 - google drive, onedrive, dropbox, etc.
 - island = 100%
 - space

cloud provider
 - Azure by Microsoft
 - AWS Amazon
 - Apple, iCloud

Shared Responsibility Model
 - Building your own data center (AWS data center)
 - Infrastructure, coded, secure it

Cloud Provider Responsibility
 - Provide data center of building, cooling, security, internet connection

Consumer Responsibility
 - You can manage your own data and provide access to right people

Shared Responsibility
 - You use cloud database (Amazon RDS) which managed by cloud provider and you have access

Types of cloud
 1. Private - Private to a specific company
 - housed on-site data center
 - IT companies
 - server room

2. Public
 - Google drive, iCloud, AWS

3. Hybrid
 - mix of public and private
 - many public

Types of cloud computing

SaaS (Software as a Service)
 - Google workspace, Microsoft 365, Zoom, Slack, Microsoft Teams

IaaS (Infrastructure as a Service)
 - Deploy and manage application (Google, AWS)

PaaS (Platform as a Service)
 - Build, test, deploy application
 - Heroku, Netlify, AWS, Azure

Consumption based Model

Capital expenditure (CapEx)
 - One-time, up-front
 - eg. buying a building, data center, company vehicle

Operational expenditure (OpEx)
 - spend on service, product over time / recurring cost
 - eg. rent, cloud service, cloud computing

pay as you go
 - storage database

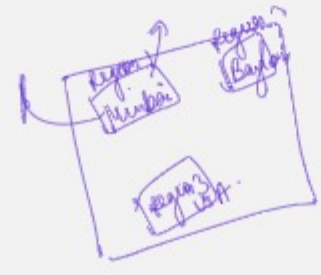
DB storage usage
 10gb → 1200
 12gb → 1250
 15gb → 1500

Time scale
 July 6 → 1200
 Aug 6 → 1500 (5gb) → 600

SaaS - you end user
 - Zoom, Slack, Google workspace

IaaS - used by developers
 - AWS, Azure, Google Cloud

PaaS - Virtual machine (Azure)
 - Google, AWS, Salesforce, Zoho, Dynamics 365



Microsoft Azure
 - cloud computing platform providing cloud based multiple services
 - AI, IoT, data storage, security, analytics, business intelligence deployment

Azure Infrastructure
 - Regions, availability zone, resources, subscriptions

Physical infra
 - extra costs
 - not managed by you
 - data center, power, cooling, network, etc.

Management infrastructure
 1. Azure resources, resource group
 - DB, VM, creating anything is a resource (single service)

2. Data center
 - you put area on earth where you have data center setup (which is physical)

3. Regions
 - you put area on earth where you have data center setup (which is physical)

4. Availability zone
 - part of region, in which if power outage, in which zone it will be available at very moment

5. Region pairs
 - Data center region are located far enough to avoid being affected together

6. Azure subscriptions
 - Azure account
 - per subscription cost subscription
 - managed group

West US, East US
 DC DC

Sovereign Region
 - special region
 - to store govt, legal, security, privacy related data

govt, legal entity